1. Linear search: read, trace, explain

Name: _____ Class: _____ Date: _____

WAGBA: Follow and explain search algorithms so we can choose the correct steps to find an item.

The library wants book number 13. Follow this early-stopping algorithm, not just where you can see 13 in the list.

items = the list below. target = the requested number. Indexes stay unchanged.

```
Index:   0   1   2   3   4   5   6
Value:   5   9  13  18  24  31  42
```

```
position ← 0
found ← False
WHILE position < LEN(items) AND found = False
    IF items[position] = target THEN
        found ← True
    ELSE
        position ← position + 1
    ENDIF
ENDWHILE
OUTPUT found
```

Read: underline the comparison, circle the instruction that advances the index, and mark the stopping condition.

Index	Value	Match?	Next action
0	5	No	Check index 1

Explain: At index __, the current value is __. This equals target __, so the algorithm __.

Correct this claim: "The first value is not the target, so output not found."

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2. Binary search: justify the boundary

Name: _____ Class: _____ Date: _____

WAGBA: Follow and explain search algorithms so we can choose the correct steps to find an item.

The list is sorted in ascending order. Search for 13. Keep original indexes. mid is an index; items[mid] is a value. items = the list below. target = the requested number. Indexes stay unchanged.

```
Index:   0   1   2   3   4   5   6
Value:   5   9  13  18  24  31  42
```

```
01  low ← 0
02  high ← LEN(items) - 1
03  found ← False
04  WHILE low ≤ high AND found = False
05      mid ← (low + high) DIV 2
06      IF items[mid] = target THEN
07          found ← True
08      ELSE
09          IF items[mid] < target THEN
10              low ← mid + 1
11          ELSE
12              high ← mid - 1
13          ENDIF
14      ENDIF
15  ENDWHILE
16  OUTPUT found
```

low	high	mid	Value	Next decision
0	6	3	18	Keep 0-2; high = 2

Explain the first change: Target __ is smaller than value __. Because the list is sorted, keep indexes __ to __. Therefore high becomes __.

Why exclude a checked midpoint when it does not match? Explain using values, not just "go left".

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3. Independent comparison and reflection

Name: _____ Class: _____ Date: _____

WAGBA: Follow and explain search algorithms so we can choose the correct steps to find an item.

Use the same list. Search for target 42 with both supplied algorithms. Do not reopen the completed model until you have tried.

items = the list below. target = the requested number. Indexes stay unchanged.

```
Index:   0   1   2   3   4   5   6
Value:   5   9  13  18  24  31  42
```

Linear: write the checked values, the result index and the number of item checks.

Binary: write each low/high/mid value and checked item, then state the result.

Explain the difference: Both algorithms __. Linear search __, whereas binary search __. Binary search needs sorted data because __.

Challenge: Does this prove binary always uses fewer checks? Find a counterexample.

My evidence and next action: I can __ independently / with __ support. Next I will __.